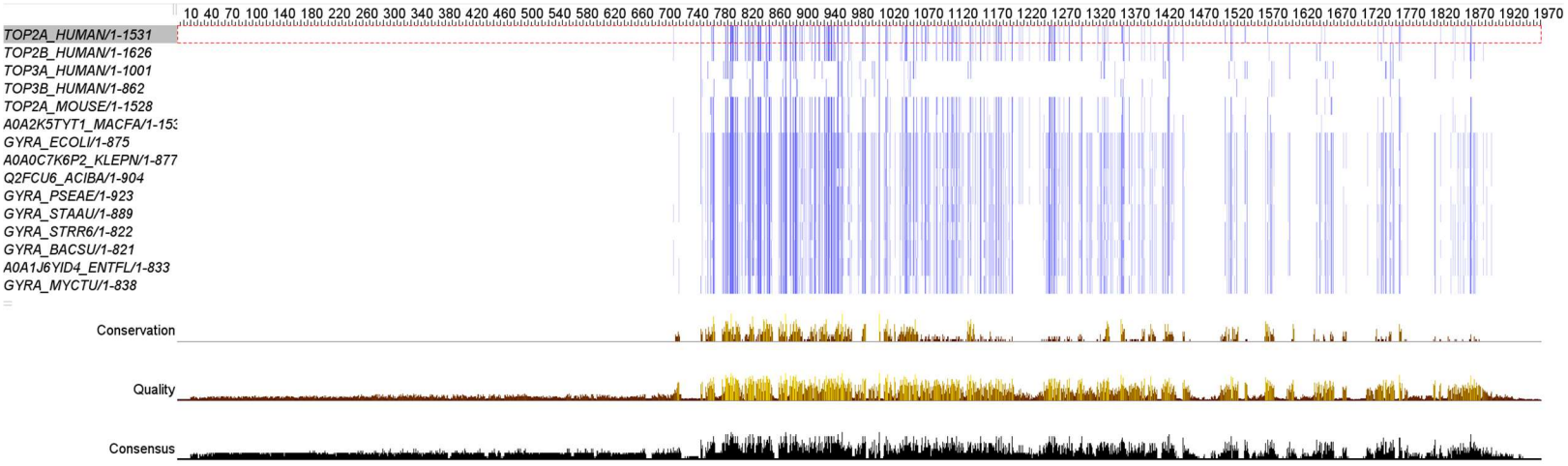
Jonathan Hummel
Homework 2
RBIF 101

Question 1:

To find type II A topoisomerase homologs, I first searched "topoisomerase" on InterPro and selected the relevant InterPro family: DNA topoisomerase, type IIA (IPR001241) [1]. Under that family entry, I searched for human proteins as a starting point and found TOPA and TOPB. I looked at the domains on InterPro under human TOP2A (P11388) to find characteristic domains that could be useful [2]. This led me to an InterPro domain called Topo_IIA_dom_A (IPR002205), whose description seemed functionally relevant and promising for finding both eukaryotic and prokaryotic type IIA homologs [3]. I then filtered the proteins under that domain's InterPro entry by each of the 12 species listed in the homework instructions. I chose homologs from among the hits by favoring sequences that were reviewed, longer, had stronger evidence, and had relevant associated PDB structures. After doing this, I noticed only 2 human paralogs with the Topo_IIA_dom_A/DNA_topoisoIV domain: TOP2A and TOP2B. This was consistent with the literature I could find on the topic [4,5]. To get at least 3 human paralogs, I went back to the InterPro entry for human TOP2A, looked at the domains section again, and found the TOPRIM domain (IPR006171) [2,6]. This led me to human TOP3A and TOP3B, which are *type IA* topoisomerases with a conserved TOPRIM domain in common with the TOP2 (i.e. type IIA) proteins [6,7]. The final table of selected sequences can be seen below with homology noted with respect to human TOP2A:

Accession	Source DB	Name	Taxological Name	Homology (to P11388)	Length	Found via	Cell type	Gram
P11388	**reviewed**	**DNA topoisomerase 2-alpha**	**Homo sapiens**	**Reference**	**1531**	**IPR002205**	**Eukarytoic**	
Q02880	reviewed	DNA topoisomerase 2-beta	Homo sapiens	Paralog	1626	IPR002205	Eukarytoic	
Q13472	reviewed	DNA topoisomerase 3-alpha	Homo sapiens	Paralog	1001	IPR006171	Eukarytoic	
O95985	reviewed	DNA topoisomerase 3-beta-1	Homo sapiens	Paralog	862	IPR006171	Eukarytoic	
Q01320	reviewed	DNA topoisomerase 2-alpha	Mus musculus	Ortholog	1528	IPR002205	Eukarytoic	
A0A2K5TYT1	unreviewed	DNA topoisomerase 2	Macaca fascicularis	Ortholog	1531	IPR002205	Eukarytoic	
P0AES4	reviewed	DNA gyrase subunit A	Escherichia coli (strain K12)	Ortholog	875	IPR002205	Prokaryotic	Negative
A0A0C7K6P2	unreviewed	DNA gyrase subunit A	Klebsiella pneumoniae	Ortholog	877	IPR002205	Prokaryotic	Negative
Q2FCU6	unreviewed	DNA gyrase subunit A	Acinetobacter baumannii	Ortholog	904	IPR002205	Prokaryotic	Negative
P48372	reviewed	DNA gyrase subunit A	Pseudomonas aeruginosa (strain PAO1)	Ortholog	923	IPR002205	Prokaryotic	Negative
P20831	reviewed	DNA gyrase subunit A	Staphylococcus aureus	Ortholog	889	IPR002205	Prokaryotic	Positive
Q8DPM2	reviewed	DNA gyrase subunit A	Streptococcus pneumoniae (strain R6)	Ortholog	822	IPR002205	Prokaryotic	Positive
P05653	reviewed	DNA gyrase subunit A	Bacillus subtilis (strain 168)	Ortholog	821	IPR002205	Prokaryotic	Positive
A0A1J6YID4	unreviewed	DNA gyrase subunit A	Enterococcus faecalis	Ortholog	833	IPR002205	Prokaryotic	Positive
P9WG47	reviewed	DNA gyrase subunit A	Mycobacterium tuberculosis (strain H37Rv)	Ortholog	838	IPR002205	Prokaryotic	Positive

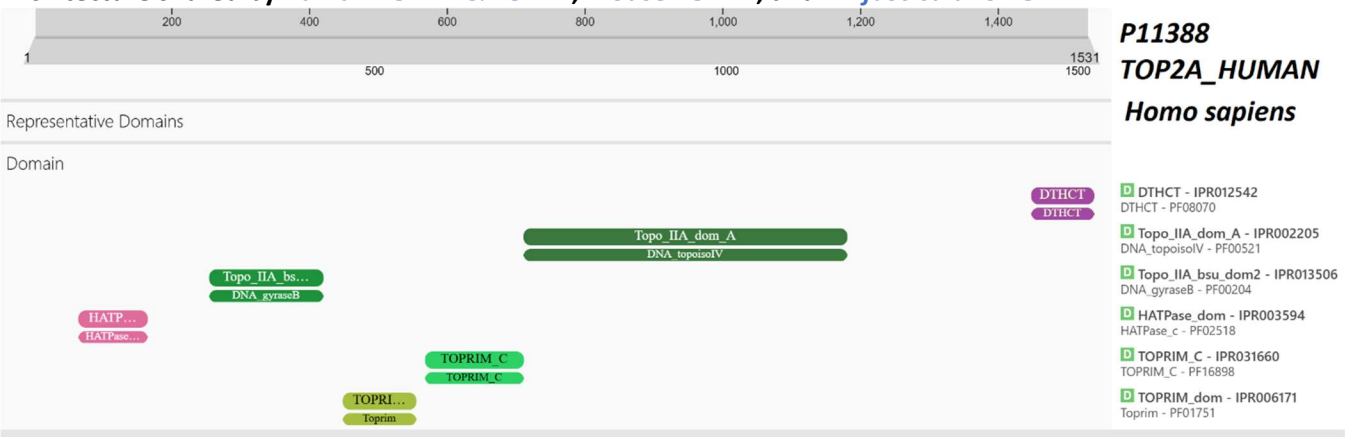
The snapshot of the MSA of these sequences in Jalview is below (aligned with ClustalΩ and colored by percent identity). For this MSA, the **pairwise percent identities range from 9.2% to 98.2% (average of 29.3%)**. The **pairwise percent similarities range from about 16.8% to 99.5% (average of 42.3%)**. As seen below, the TOP3 sequences are responsible for the lowest end of those identity/similarity ranges, which makes sense because they only have one much smaller conserved domain (TOPRIM) in common with the TOP2 proteins. To calculate the percentage identity and similarity ranges reported above, I used the ClustalΩ command line tool to get the percent identity matrix, along with a web service (SIAS) for calculating similarity matrices from MSAs [8,9].



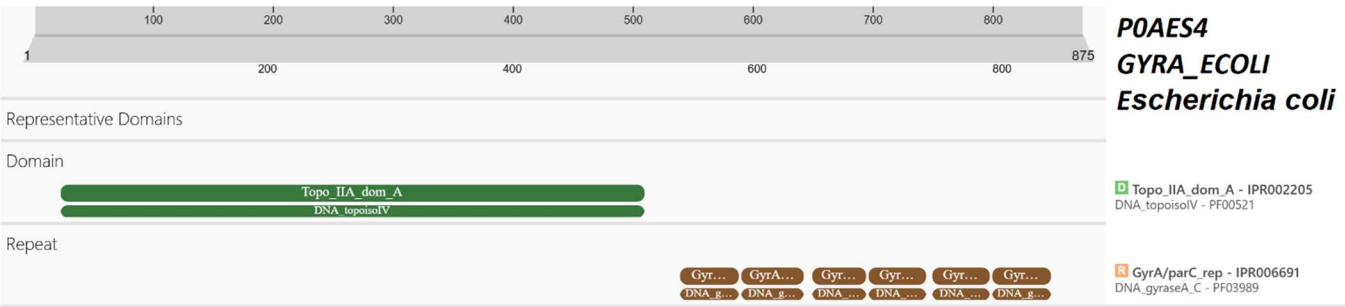
Question 2:

The snapshots that follow are from InterPro and were obtained by running a search for Pfam domains in all sequences represented in the MSA shown earlier. After looking at the domain architecture images generated for all sequences, the Pfam domain architecture from only one representative sequence was chosen and displayed below. All sequences/species that had the same architecture (domains and order) are noted above each image. The color of the DTHCT domain was changed for easier distinction from the GyrA/parC repeat domain.

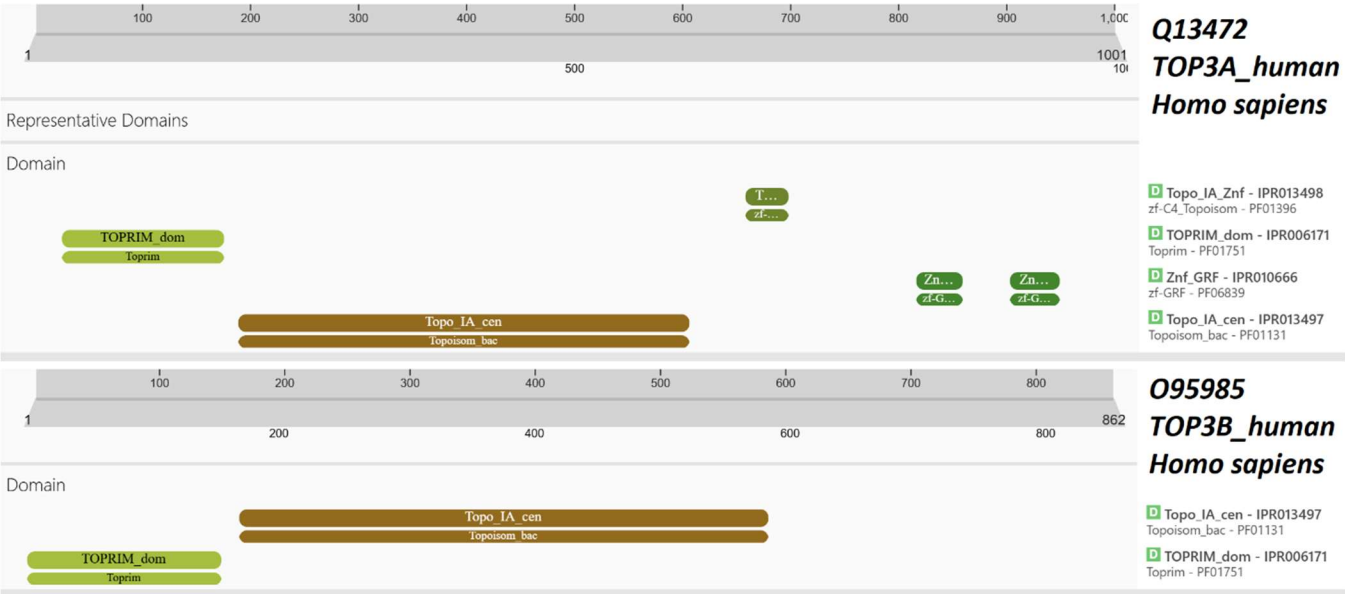
Architecture shared by human TOP2A & TOP2B, mouse TOP2A, and *M. fascicularis* TOP2A:



Architecture shared by gyrA sequences from *E. coli*, *K. pneumoniae*, *A. baumannii*, *P. aeruginosa*, *S. aureus*, *S. pneumoniae*, *B. subtilis*, *E. faecalis*, and *M. tuberculosis*:



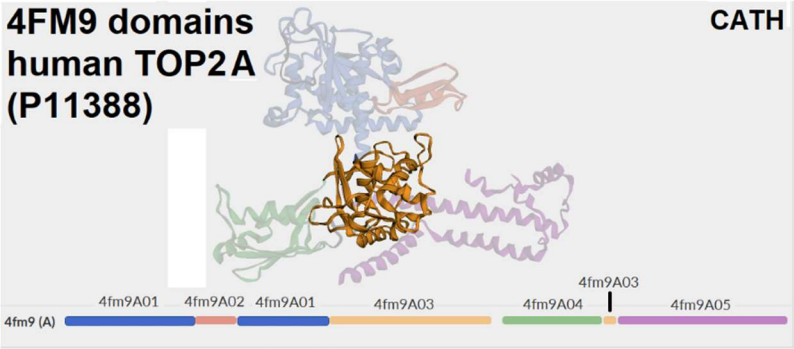
The human TOP3 proteins have different architectures, but both include TOPRIM_dom and Topo_IA_cen:

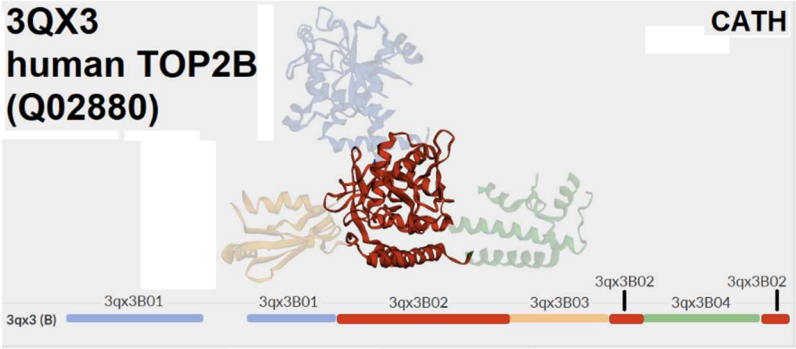


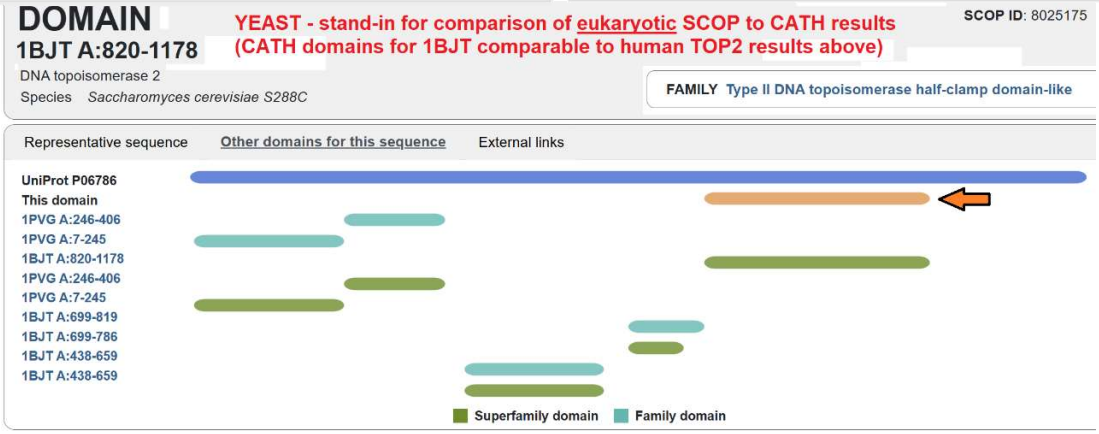
Question 3:

I chose PDB structures for human TOP2A (4FM9), human TOP2B (3QX3), *E. coli* gyrA (1AB4), and *B. subtilis* gyrA (4DDQ). The domain architectures I was able to obtain for these structures using CATH and/or SCOP2 can be seen below. Due to limited SCOP results, the SCOP architecture for a structure from yeast is included for comparison to the human CATH results, as that structure's CATH architecture was comparable to the human ones shown below. Snapshots were edited to be more compact than the original CATH and SCOP entries.

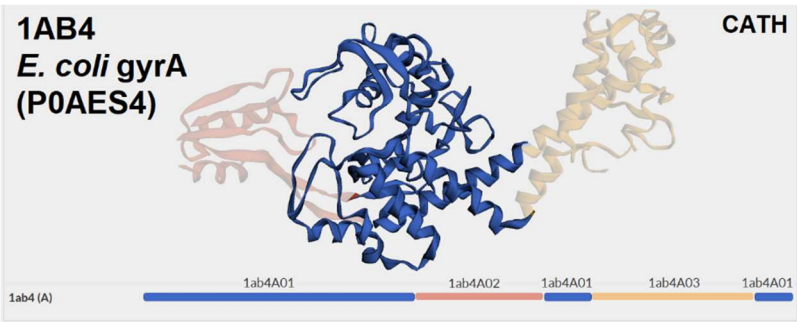
Eukaryotic structures and architectures:

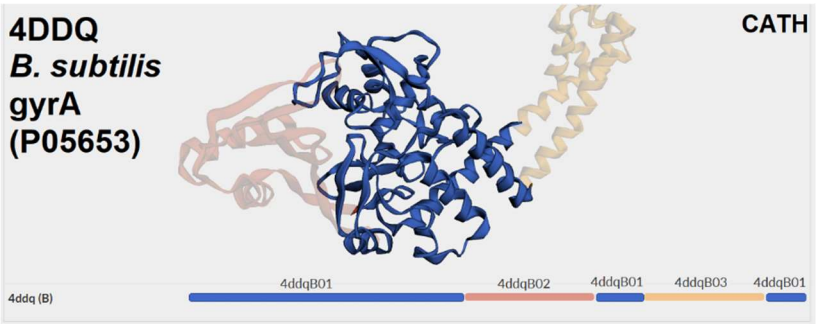


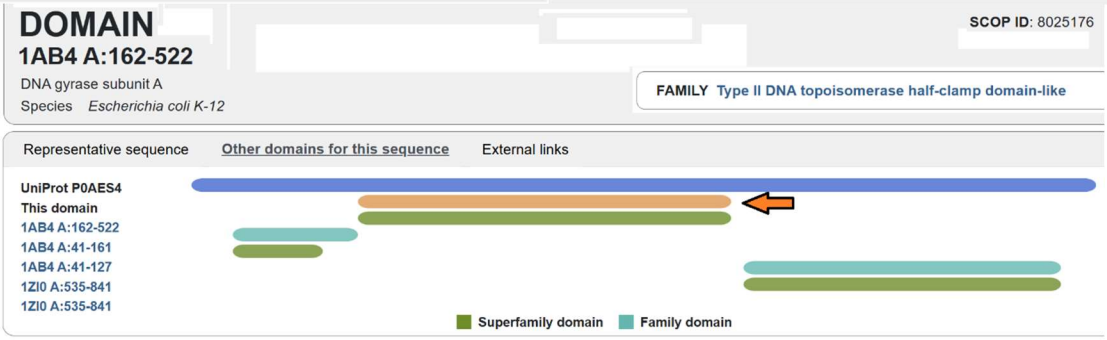




Prokaryotic structures and architectures (4DDQ shown for reference later, despite identical architecture):





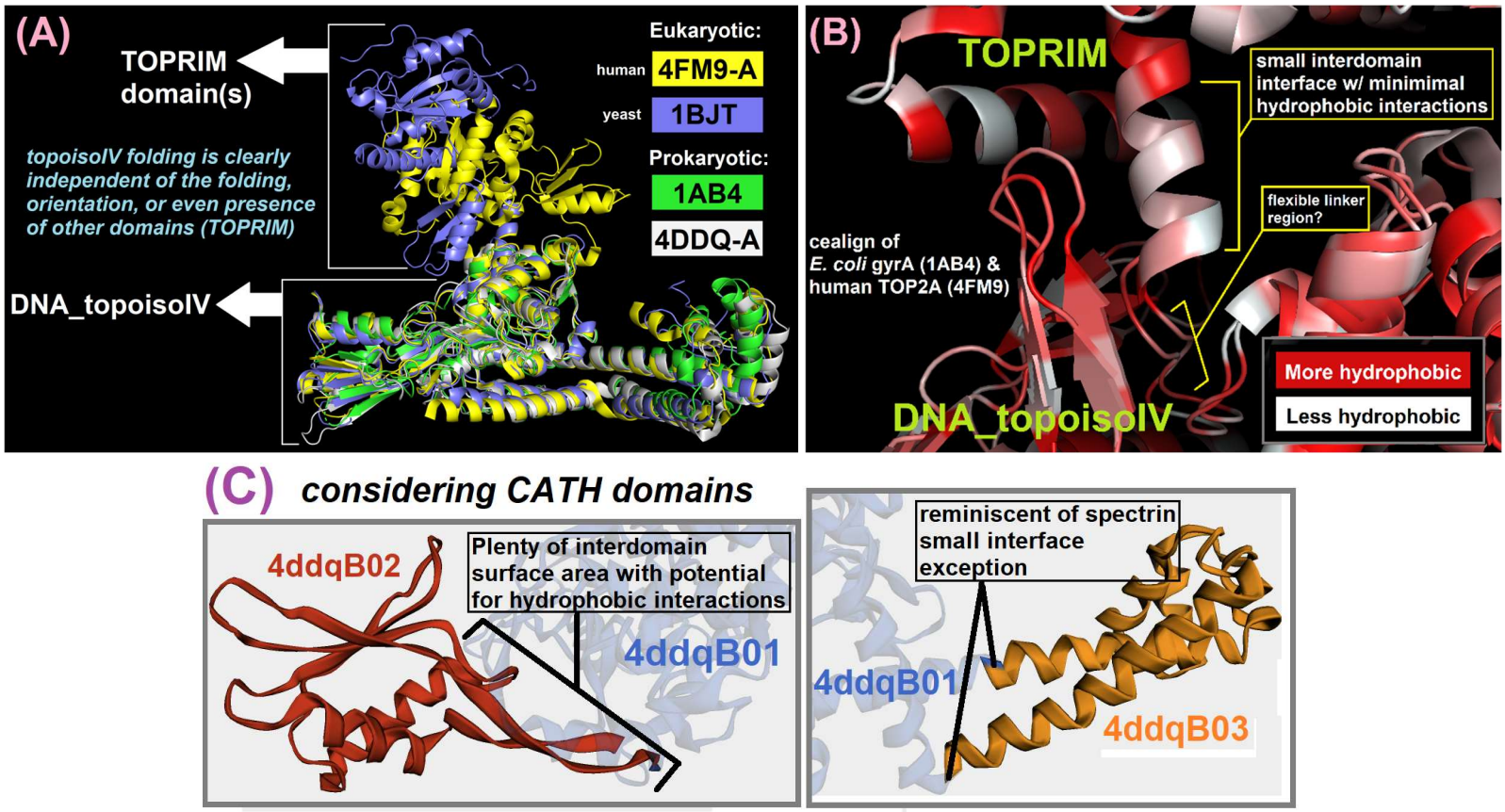


There are certainly some differences between the SCOP, CATH, and Pfam results. One major difference between CATH and SCOP for both the prokaryotic and eukaryotic structures is that CATH makes more structural domain divisions. CATH divides the sequence corresponding to the Pfam DNA_topoisoIV domain into 3 structural domains, while SCOP classifies the same region as one domain (see orange arrows and bars in SCOP results). This is consistent with the observation made in some of the Week 3 reading materials that CATH tends to cut structures into more domains than SCOP [10]. For this same reason, CATH's domain designation differs the most from Pfam, which assigns that entire region as one domain (DNA_topoisoIV). Given the strong structural conservation of the DNA_topoisoIV domain seen on the previous page, it likely folds independently as one unit in all the homologous sequences. For this reason, SCOP's (and Pfam's) classification of it as one domain may make the most sense. It is worth noting that SCOP and Pfam still designate domains differently, despite being more similar to each other than CATH. CATH's structural domain designation is also apparently more complex than SCOP's because it allows domains that are non-contiguous within the sequence. This behavior of CATH is also different from how Pfam assigns domains. This discontinuity of CATH's structural domains is the main reason that structures for human TOP2A and TOP2B had slightly different architectures, whereas they had the same architecture on Pfam (and likely would have on SCOP, given the yeast proxy results). A major difference between the CATH and Pfam results arises because PDB structures do not always capture the full sequence. For example, the CATH results for the human TOP2 structures do not include domains corresponding to the HATPase, DNA_gyraseB, or DTHCT domains from the eukaryotic Pfam architectures earlier because they were not captured in these structures (according to PDBsum) [11,12]. A similar observation can be made about the DNA_gyraseA_C domain of the prokaryotic sequences shown from Pfam earlier, which is also not represented in the CATH results due to its absence from the PDB structures [13,14]. A major difference between CATH and SCOP is their comprehensiveness with regard to PDB structures. *E. coli* gyrA was the only protein/sequence from Question 1 that had a structure with both the DNA_topoisoIV domain included and any SCOP annotation. I made a point to choose UniProt sequences that had many relevant PDB structures earlier (checked via PDBsum), and I tried many types of searches on SCOP with limited results (SCOP2 and SCOPe). This is why I resorted to a structure from yeast with the DNA_topoisoIV domain for comparison to the human CATH results. Meanwhile, CATH had entries and domain annotations for many of the PDB structures I searched for.

Question 4:

In their paper, *Han et al.* essentially said that domains within a protein tend to fold *independently* when their connecting region is sufficiently small and/or flexible, while domains often fold *dependently* when there is more extensive interdomain surface area at the interface (often hydrophobic) [15]. While not definitive, I see quite a bit of overlap and consistency with the findings by *Han et al.* in the results from Question 3. **Figures A**, **B**, and **C** on the next page were made to illustrate the consistencies that I am about to describe. First, the prokaryotic sequences/structures from Question 3 do not include TOPRIM domains, which supports the independence of DNA_topoisoIV's folding. In **Figure A**, representative eukaryotic and prokaryotic structures were aligned using *cealign* in PyMol. After doing this alignment, it is clear that the DNA_topoisoIV folding is highly consistent in all homologs regardless of whether another domain is present (e.g. TOPRIM for the eukaryotes). Between the yeast and human structures, there is a notable deviation in the folding and positioning of TOPRIM, but the DNA_topoisoIV domain's folding remains extremely similar in both. Given that these domains fold independently, is worth looking at the region that connects them. This is done in **Figure B**, in which it can be seen that the structure connecting TOPRIM and DNA_topoisoIV is small, has minimal hydrophobic interface surface area, and might even be somewhat flexible due to a less ordered region under TOPRIM's connecting α-helix. Because of those details, this interface does seem to have more in common with those listed under "Proteins that show independent domain folding" in Table 2 from *Han et al.* [15]. Also, while I think the SCOP classification of DNA_topoisoIV as one domain makes more sense (it clearly folds independently), I can also see themes discussed by *Han et al.* if I consider CATH's domain assignments from Question 3 [15]. **Figure C** highlights how the CATH domain interfaces have qualities that are consistent with dependent folding according to *Han et al.* [5]. One of the domain interfaces has extensive surface area, likely with plenty of opportunities for hydrophobic interactions at the interface (**Figure C**, left). The other interface is small but similar to the "Spectrin

exception" discussed in the paper because the α-helices extend from one CATH domain into another, making their folding dependent (**Figure C**, right) [15].



Question 5:

5(a): *sequence-based domain annotation tool (InterPro/Pfam)*

Accession	Gene	Name	Tax Name	Length	Entry Accession
A4D2B8	PMS2P1	Putative postmeiotic segregation increased 2-like protein 1	Homo sapiens	440	IPR036890
A6NHR9	SMCHD1	Structural maintenance of chromosomes flexible hinge domain-containing protein 1	Homo sapiens	2005	IPR036890
A8MQ11	PMS2P5	Postmeiotic segregation increased 2-like protein 5	Homo sapiens	134	IPR036890
O14874	BCKDK	Branched-chain alpha-ketoacid dehydrogenase kinase	Homo sapiens	412	IPR036890
O95744	PMS2P2	Putative postmeiotic segregation increased 2-like protein 2	Homo sapiens	297	IPR036890
P07900	HSP90AA1	Heat shock protein HSP 90-alpha	Homo sapiens	732	IPR036890
P08238	HSP90AB1	Heat shock protein HSP 90-beta	Homo sapiens	724	IPR036890
P11388	TOP2A	DNA topoisomerase 2-alpha	Homo sapiens	1531	IPR036890
P14625	HSP90B1	Endoplasmin	Homo sapiens	803	IPR036890
P40692	MLH1	DNA mismatch repair protein Mlh1	Homo sapiens	756	IPR036890
P54277	PMS1	PMS1 protein homolog 1	Homo sapiens	932	IPR036890
P54278	PMS2	Mismatch repair endonuclease PMS2	Homo sapiens	862	IPR036890
Q02880	TOP2B	DNA topoisomerase 2-beta	Homo sapiens	1626	IPR036890
Q12931	TRAP1	Heat shock protein 75 kDa, mitochondrial	Homo sapiens	704	IPR036890
Q13401	PMS2P3	Putative postmeiotic segregation increased 2-like protein 3	Homo sapiens	168	IPR036890
Q13670	PMS2P11	Putative postmeiotic segregation increased 2-like protein 11	Homo sapiens	270	IPR036890
Q14149	MORC3	MORC family CW-type zinc finger protein 3	Homo sapiens	939	IPR036890
Q14568	HSP90AA2P	Heat shock protein HSP 90-alpha A2	Homo sapiens	343	IPR036890
Q15118	PDK1	[Pyruvate dehydrogenase (acetyl-transferring)] kinase isozyme 1, mitochondrial	Homo sapiens	436	IPR036890
Q15119	PDK2	[Pyruvate dehydrogenase (acetyl-transferring)] kinase isozyme 2, mitochondrial	Homo sapiens	407	IPR036890
Q15120	PDK3	[Pyruvate dehydrogenase (acetyl-transferring)] kinase isozyme 3, mitochondrial	Homo sapiens	406	IPR036890
Q16654	PDK4	[Pyruvate dehydrogenase (acetyl-transferring)] kinase isozyme 4, mitochondrial	Homo sapiens	411	IPR036890
Q58FF6	HSP90AB4P	Putative heat shock protein HSP 90-beta 4	Homo sapiens	505	IPR036890
Q58FF7	HSP90AB3P	Putative heat shock protein HSP 90-beta-3	Homo sapiens	597	IPR036890
Q58FF8	HSP90AB2P	Putative heat shock protein HSP 90-beta 2	Homo sapiens	381	IPR036890
Q58FG0	HSP90AA5P	Putative heat shock protein HSP 90-alpha A5	Homo sapiens	334	IPR036890
Q58FG1	HSP90AA4P	Putative heat shock protein HSP 90-alpha A4	Homo sapiens	418	IPR036890
Q86VD1	MORC1	MORC family CW-type zinc finger protein 1	Homo sapiens	984	IPR036890
Q8TE76	MORC4	MORC family CW-type zinc finger protein 4	Homo sapiens	937	IPR036890
Q9NZJ4	SACS	Sacsin	Homo sapiens	4579	IPR036890
Q9UHC1	MLH3	DNA mismatch repair protein Mlh3	Homo sapiens	1453	IPR036890
Q9Y6X9	MORC2	ATPase MORC2	Homo sapiens	1032	IPR036890

To get the homology hits shown on the previous page, I first searched InterPro for an *E. coli* gyrase subunit B sequence (UniProt: P0AES6) [16]. I went to the domain architecture section of the InterPro page for that protein, saw the HATPase domain (IPR003594), and clicked on the homologous superfamily it fell under (IPR036890) [17]. I went to the "Proteins" section of the Histidine kinase/HSP90-like ATPase superfamily and then filtered by "Homo sapiens" and reviewed sources to keep the results manageable and relevant. This yielded 32 hits, all of which can be seen in the table on the previous page. If I had a sequence that was not in UniProt, I could have followed virtually the same steps just described, the only difference being an initial search by sequence on InterPro to get the domain annotations (and detect the HATPase domain, if present) [18].

5(b): *structural domain annotation tool (SCOP2)*

SCOP ID	PDB	Chain:Res	UniProt	Gene Name	Residues	SCOP Hierarchy (TP=1,CL=1000003,CF=2000066,...)	Protein Name	Organism
8020090	**1Y8O**	**A:177-301**	**Q15120**	**PDK3**	**177-301**	**SF=3000091,FA=4003507**	**[Pyruvate dehydrogenase (acetyl-transferring)] kinase isozyme 3, mitochondrial**	**Homo sapiens**
8023547	1GKZ	A:186-378	Q00972	Bckdk	216-408	SF=3000091,FA=4003507	Branched-chain alpha-ketoacid dehydrogenase kinase	Rattus norvegicus
8023964	**1H7S**	**A:29-231**	**P54278**	**PMS2**	**29-231**	**SF=3000091,FA=4000168**	**Mismatch repair endonuclease PMS2**	**Homo sapiens**
8024776	1JM6	A:1177-1366	Q64536	Pdk2	185-374	SF=3000091,FA=4003507	[Pyruvate dehydrogenase (acetyl-transferring)] kinase isozyme 2, mitochondrial	Rattus norvegicus
8028628	1UYL	A:17-224	P07900	HSP90AA1	17-224	SF=3000091,FA=4001945	Heat shock protein HSP 90-alpha	Homo sapiens
8029126	2O1U	A:85-337	P41148	HSP90B1	85-337	SF=3000091,FA=4001945	Endoplasmin	Canis lupus familiaris
8070951	1UYM	A:16-226	P08238	Os09g0482600	11-221	SF=3000091,FA=4001945	Heat shock protein HSP 90-beta	Homo sapiens
8070969	5F3K	A:82-294	Q12931	htpG	82-294	SF=3000091,FA=4001945	Heat shock protein 75 kDa, mitochondrial	Homo sapiens
8098513	4NH9	A:73-337	P14625	HSP90B1	73-337	SF=3000091,FA=4001945	Endoplasmin	Homo sapiens

For this part, I went to the SCOP2 website and searched for the same UniProt accession for gyrB used in **5(a)** (P0AES6) [16,19]. This led me to the SCOP domain represented by 1EI1 A:2-220 (8023824), which is an *E. coli* gyrB structure [20]. I checked that this structural domain included the HATPase domain using PDBsum [21]. From there I clicked "Show ancestry" and went to the SCOP superfamily for this domain: GHKL (Gyrase, Hsp90, Histidine Kinase, MutL) (3000091) [22]. The family names under that superfamily looked promising, so I downloaded one of the SCOP parsable files and used *grep* to extract all structures under the superfamily 3000091. This yielded 55 new representative structure hits in total [22]. I annotated the list of hits using a Python script I wrote that retrieves UniProt accessions and descriptive information for PDB IDs (if possible). To keep the SCOP results manageable and relevant, I filtered the list to include only structures from mammals using MS Excel (shown above). The **bolded** hits are aligned to the ATPase domain of *E. coli* gyrB at the end.

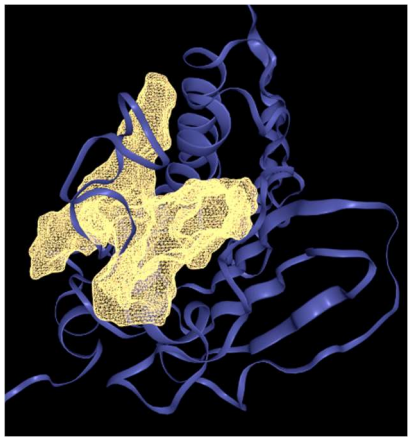
5(c): *structural alignment tool (VAST)*

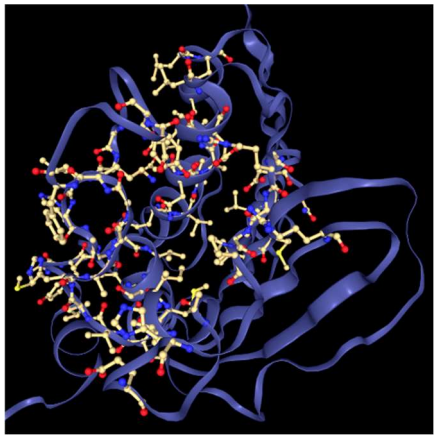
PDB C D	Alignment Length	SCORE	RMSD	Identity	UniProt	Protein Name	Organism
4R1F C	**215**	**20.48**	**2.4**	**23.3**	**P11388**	**DNA topoisomerase 2-alpha**	**Homo sapiens**
1H7S A	145	17.1	2.78	19.3	P54278	Mismatch repair endonuclease PMS2	Homo sapiens
6ZY7 B	169	16.64	1.96	23.7	P11388	DNA topoisomerase 2-alpha	Homo sapiens
6MW7 A	147	15.74	2.68	14.3	A6NHR9	Structural maintenance of chromosomes flexible hinge domain-containing protein 1	Homo sapiens
7C7B A	159	15.27	3.11	14.5	Q12931	Heat shock protein 75 kDa, mitochondrial	Homo sapiens
6O1E A	151	15.11	3.01	18.5	Q14149	MORC family CW-type zinc finger protein 3	Homo sapiens
5OFB B	**160**	**14.85**	**3.16**	**15**	**Q9Y6X9**	**ATPase MORC2**	**Homo sapiens**
5HPH B	175	14.72	3.2	14.9	Q12931	Heat shock protein 75 kDa, mitochondrial	Homo sapiens
1OSF A	135	13.62	2.75	14.1	P07900	Heat shock protein HSP 90-alpha	Homo sapiens
5V46 A	146	12.97	3.63	12.3	Q9NZJ4	Sacsin	Homo sapiens
3CRL B	98	10.83	1.86	22.4	P10515	Dihydrolipoyllysine-residue acetyltransferase component	Homo sapiens
8EGF A	88	10.15	1.84	22.7	Q00972	Branched-chain alpha-ketoacid dehydrogenase kinase	Rattus norvegicus
6W3I B	46	7.49	2.04	10.9	Q5VWP2	Terminal nucleotidyltransferase 5C	Homo sapiens
6TM2 B	34	7.12	1.83	5.9	Q02817	Mucin-2	Homo sapiens
8A43 A	48	6.85	2.58	12.5	Q9P1U0	DNA-directed RNA polymerase I subunit RPA12	Homo sapiens
4Z2M B	38	6.42	2.15	13.2	Q9Y5B9	FACT complex subunit SPT16	Homo sapiens
7KRJ C	38	6.08	2.02	2.6	Q15185	Prostaglandin E synthase 3	Homo sapiens

For this part, I went to the VAST Search tool hosted by NCBI, entered the PDB code for the *E. coli* gyrB structure found on SCOP in **5(b)** (1EI1), and then started a search against the medium redundancy PDB subset [23]. VAST

detected 2 chains, each with 2 domains. I compared them to the entry for 1EI1 on PDBsum and saw that the chains are identical and domain 1 detected by VAST overlapped best with HATPase [21]. I selected the results for domain 1 from 1EI1-A, copied the table of 59 new representative hits from VAST, and annotated it with information from UniProt using the same Python script I used in **5(b)**. I then sorted the annotated list to include only structures from mammals to get the list of hits on the previous page, which is ordered by decreasing alignment score (sorted/ordered in Excel). Once again, **bolded** structures are used in the alignments at the end.

5(d): *binding site comparison tool (DoGSiteSCorer + SIENA)*





PDB code	PDB chains	Ligand PDB code	Active site identity	# Active site mutations	# Active site gaps	Backbone RMSD	UniProt	Protein Name	Organism
3G7E	A	B46_A_1	0.981818	1	0	3.511669	P0AES6	DNA gyrase subunit B	Escherichia coli (strain K12)
7P2W	A	4QR_A_301	0.963636	1	1	5.84082	P0AES6	DNA gyrase subunit B	Escherichia coli (strain K12)
7DQM	A		0.872727	1	6	3.611694	P0AES6	DNA gyrase subunit B	Escherichia coli (strain K12)
5MMN	A	O54_A_301	0.836364	1	8	3.638828	P0AES6	DNA gyrase subunit B	Escherichia coli (strain K12)
7C7O	A	FKU_A_301	0.8	1	10	2.101561	P0AES6	DNA gyrase subunit B	Escherichia coli (strain K12)
6M1J	A	EZ6_A_305	0.763636	4	9	3.563463	Q9I7C2	DNA gyrase subunit B	Pseudomonas aeruginosa (strain PAO1...)
9GGQ	*B*		*0.690909*	*1*	*16*	*1.375065*	*A0A140N6C1*	*DNA gyrase subunit B*	*Escherichia coli BL21(DE3)*
7PQL	*A*	*81N_A_302*	*0.636364*	*5*	*15*	*0.348007*	*Q43907*	*DNA gyrase subunit B*	*Acinetobacter baumannii*
7PTG	**A**	**80S_A_301**	**0.618182**	**4**	**17**	**0.645359**	**Q9I7C2**	**DNA gyrase subunit B**	**Pseudomonas aeruginosa (strain PAO1...)**
8QQS	D		0.509091	1	26	1.131299	P0AES6	DNA gyrase subunit B	Escherichia coli (strain K12)

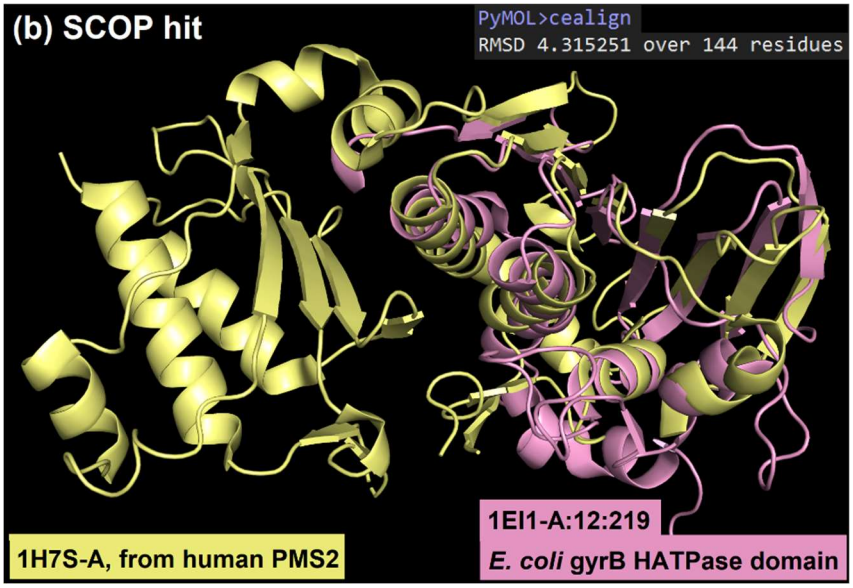
I initially struggled to find a functioning tool from Table 1 in the chapter by *Xie and Hwang* on ligand binding site detection that enabled a query for proteins with comparable binding sites [24]. That said, classmates in the Week 4 discussion found where DoGSiteSCorer is currently being hosted and a tool called SIENA is hosted in the same place, which enables queries based on the detected binding pockets or even ligands [24,25]. Thus, I went to Proteins.plus and uploaded a PDB file containing only the HATPase domain of *E. coli* Gyrase B. This was prepared in PyMol from 1EI1 by extracting and exporting residues 12-219 of Chain A, which was determined using PDBsum (CATH domain 1 covers Pfam HATPase) [21,26]. I then went to the DoGSiteSCorer tool and selected "Pockets" for analysis, "Properties" for granularity, and "All" for chains [27]. Several binding pockets were detected, but as stated in the reading material, the largest pocket (~1289 $Å^3$) was the one of most interest (i.e. ATP binding pocket). I confirmed this from the PDB entry for 1EI1, which has a bound competitive inhibitor quite similar in shape and size to ATP [26]. By choosing this pocket, I am assuming we are interested in the risk/unintended targets associated with a competitive inhibitor targeting the ATPase domain, rather than an allosteric one. I loaded the detected APT pocket (P_0) into SIENA and selected the "Flexibility analysis", but changed some of the settings in the expert mode. One thing I noticed with SIENA is that no matter which analysis mode I chose, or how lenient I made the comparison settings (e.g. min binding site identity, flexibility sensitivity, etc.), this tool only returned hits from other bacterial gyrase structures. I will discuss why I think this happened later. The table of results above was obtained by using the most lenient settings that did not yield an error in the SIENA flexibility analysis: minimum fragment length = 15, flexibility sensitivity = 1, max fragment distance = 7, minimum site identity = 0.3. This yielded 32 hits, but there were many redundant structures among these hits. I

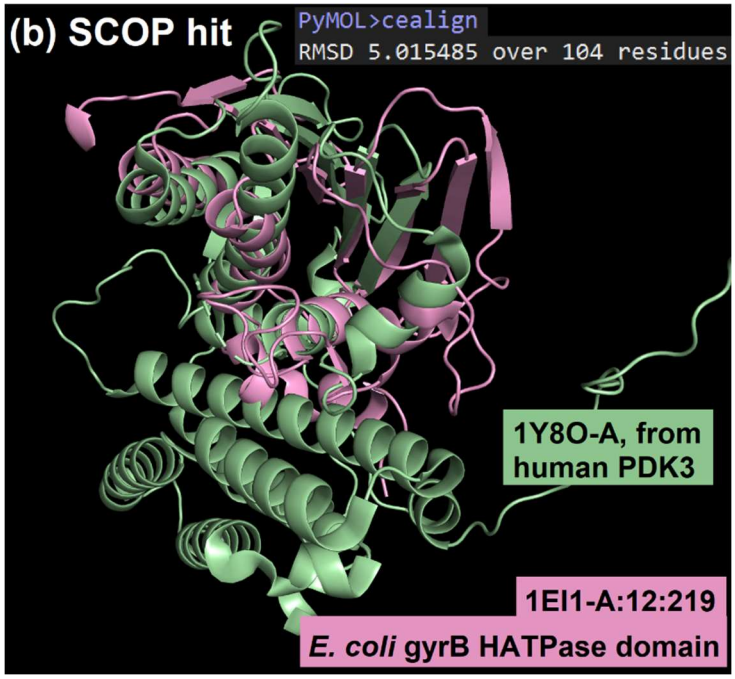
reran the SIENA flexibility query with the same settings, but with these additional measures to reduce redundancy: “remove sequence duplicates”, “remove sites with ligand duplicates”, and “ensemble size for backbone clustering” set to 10. This made for a less redundant list of 10 hits that still represented all unique UniProt sequences as the original list of 32 hits. I annotated this list with UniProt information using the Python script mentioned in **5(b)** and **5(c),** and these results can be seen on the previous page. As mentioned, the only hits I could get with DoGSiteSCorer + SIENA were other bacterial gyrases, regardless of the modes/settings used. As such, the results on the previous page are perhaps the least useful for a homology search aiming to uncover potential unintended targets in important human enzymes for compound(s) made to target the ATPase domain of bacterial gyrase. The **bolded** hits in that table are used in the structural alignments at the end and the *italicized* hits were annotated manually by looking at the PDB and UniProt (*blue* indicates what was added):

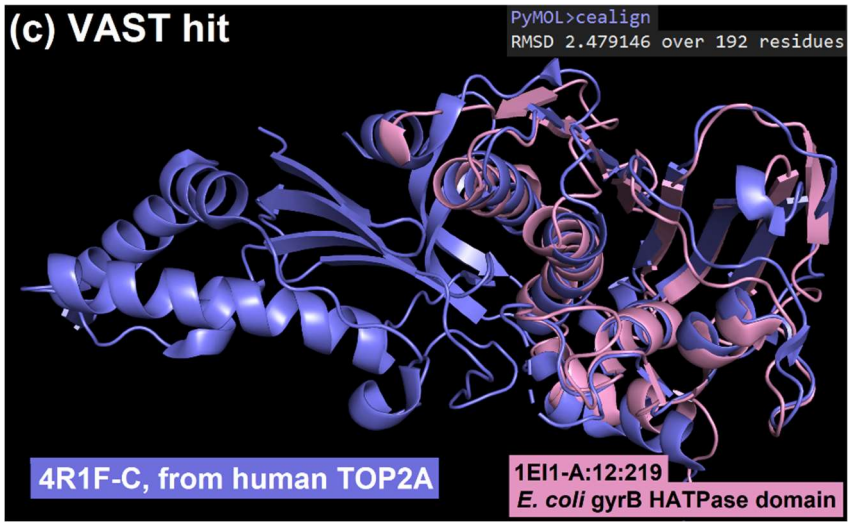
Differences in 5(a) - (d): The list of human homologs found via InterPro/Pfam in **5(a)** is the longest and most comprehensive. It covers many unique human proteins that contain an ATPase domain homologous to that of *E. coli* Gyrase B. By comparing protein descriptions and UniProt accessions, I can see qualitatively that the InterPro list from **5(a)** covers everything contained in the much smaller list from SCOP in **5(b)**. The list from VAST in **5(c)** still has plenty of overlap with the first two lists, but it also has several hits not found in the results from InterPro/Pfam, and apparent examples include Terminal nucleotidyltransferase 5C, DNA-directed RNA polymerase I subunit RPA12, FACT complex subunit SPT16, and Mucin-2. Finally, the list of hits obtained in **5(d)** by identifying the ligand binding site for the ATPase domain of gryB using DoGSiteSCorer, and then searching for proteins with comparable pockets using SIENA, was the least similar to the others. As mentioned previously, no matter what mode I used with SIENA (or how lenient the matching criteria) the only hits I got by ligand binding site comparison were from other bacterial gyrB structures. As such, this list had zero overlap with any of the first 3 lists and was perhaps the least useful for identifying potential unintended targets in eukaryotes/humans for a competitive inhibitor of the HATPase domain in bacteria.

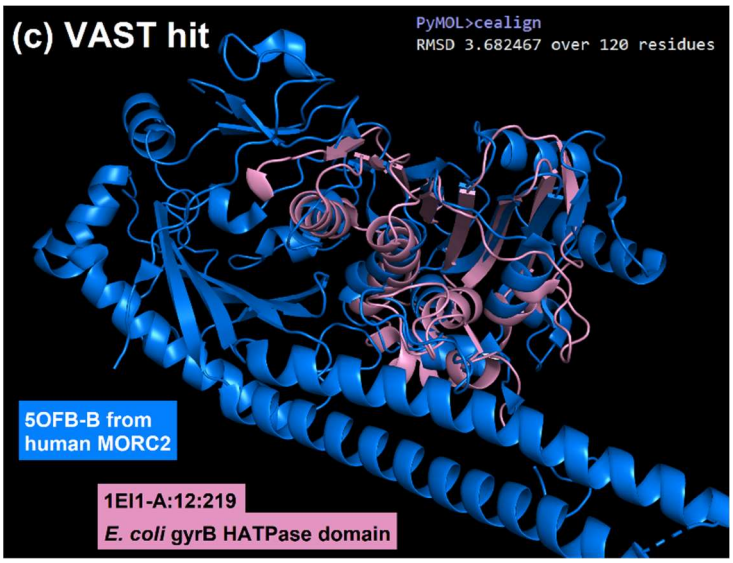
I think the differences between these lists make sense. While structure-based methods theoretically have the potential to unearth more distant or previously unrecognized homology via the ATPase domain’s conserved *structure*, I am not surprised that InterPro/Pfam was the most fruitful. The annotations on InterPro represent a large portion of established human knowledge about the domains, functional entities, and/or homology of proteins. Thus, it *if* you can confidently annotate a section of sequence with “HATPase” via Pfam, it will unlock a large number of homologs in any well-studied species. It makes sense that SCOP had much fewer hits because the list of proteins represented in SCOP2 is probably much smaller and somewhat out-of-date compared to InterPro. Also, just as importantly, SCOP aims to express different levels of structural hierarchy via representative structures, so PDB structures are clustered. For example, human TOP2A is not in my SCOP results shown earlier, but this is simply because the TOP2 domains are represented by structures from yeast (8026786) [28]. For this reason, I should have probably filtered SCOP hits by eukaryotes rather than mammals, which would have yielded ~23 hits instead of 9. A structural alignment search via VAST yielded the most unique hits compared to InterPro. This could again have to do with the different sets being searched against, as I imagine the proteins represented in the medium redundancy PDB subset and Pfam may not overlap perfectly. That said, structural alignment involving the HATPase domain is a fundamentally different way to search for homologs compared to InterPro because hits are found by comparing the type, direction, and connectivity of secondary structural elements in the query’s 3D structure to that of other proteins, as per the lecture notes. This method does not depend on sequence identity to find similar proteins. Thus, it makes sense that VAST found some hits that were not found via InterPro, examples of which were mentioned above. Supporting this is the fact that most of the hits unique to VAST shown above had very low sequence identity (<15%) with the HATPase domain from *E. coli* gyrB. Perhaps the biggest surprise for me was the fact that the ligand binding site comparison approach yielded the least useful or similar results when compared to the other methods. This may have been due to my misunderstanding of the tool, but I tried many settings/options to see if I could get more interesting hits than a list of more gyrB structures. In hindsight, it may make sense that this method identified only gyrB structures. As seen in the images on the previous page, the pocket that DoGSiteSCorer detected was large (relative to the HAPTase

domain) and irregularly shaped (almost like an "H"). The residues implicated in the binding pocket by this geometric ligand binding site detection method together span one entire side of the domain. Viewed in this way, DoGSiteSCorer identified a pocket that was pretty distinctive in shape and sequence. I do not know the inner workings of how SIENA uses this information to conduct a search, but it stands to reason that SIENA was working with a pocket that acted like a fairly unique "fingerprint", making the identification of more distant and interesting proteins impossible. That said, it may have simply been a user error on my part. I think the concept of searching by structural similarity via the ligand binding site alone sounds like it has the potential to yield the most relevant results in an assessment of potential off-target risk, e.g. for a competitive inhibitor targeting bacterial gyrB's HATPase domain. That said, I would be hesitant to assume that because the SIENA results earlier did not yield hits in mammals (or even eukaryotes), off-target effects involving important human enzymes are not possible for an antibacterial compound like the one just described. The first three methods identified a comprehensive set of human proteins that have an ATPase domain that is similar in sequence and/or structure to the one in bacterial gyrases (i.e. homologs). Based on my first glance/impression, many of these human proteins perform essential functions (e.g. PDK enzymes) so the potential off-target effects may be far from acceptable! The structural alignments between the HATPase domain of *E. coli* gyrB and some of the hits from **5(b) – (d)** can be seen below and on the next page. The images of those alignments look down over the mouth of the HATPase binding pocket, and it can be seen that this pocket appears to be quite similar (at least structurally) across a broad range of important human proteins (including TOP2A):

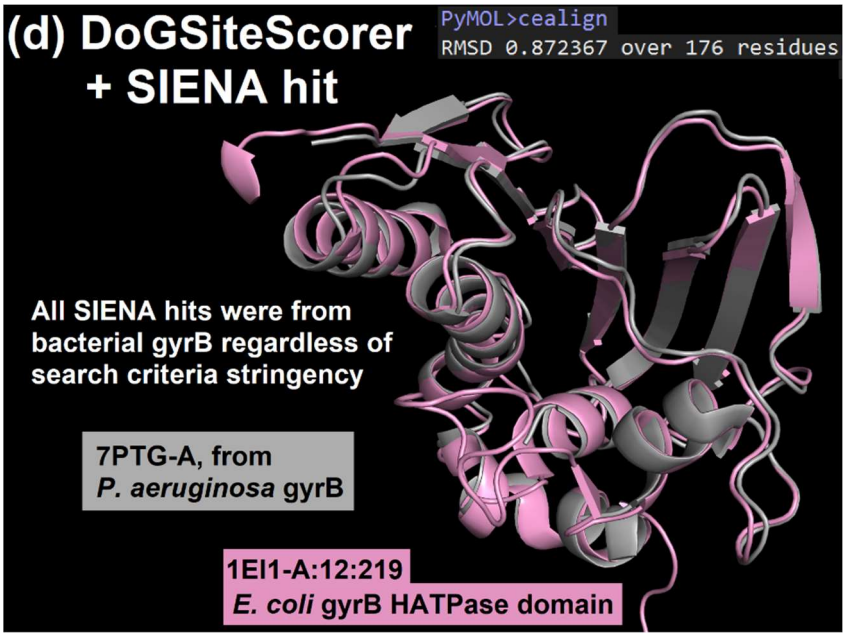








Alignment with structure from 5(d) on next page…



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